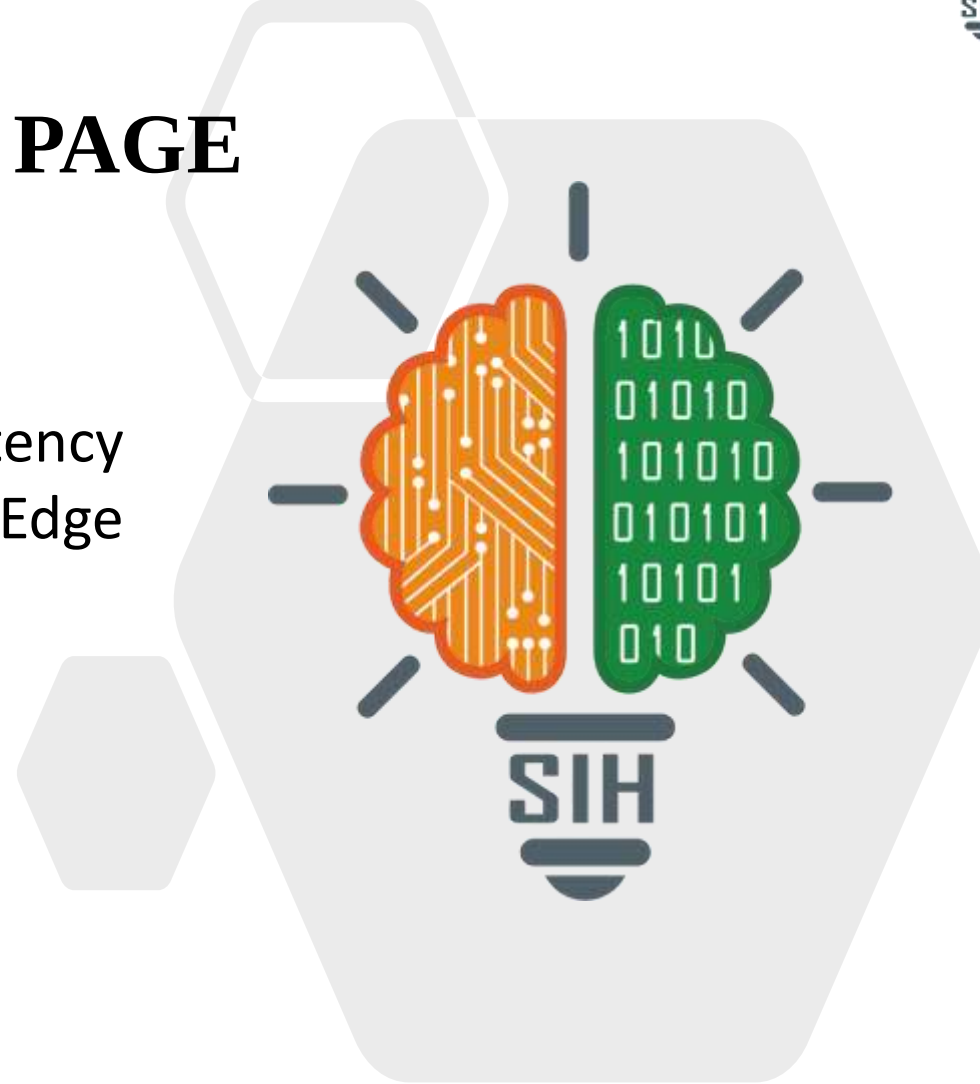


TITLE PAGE

- **Problem Statement ID** –SIH26172
- **Problem Statement Title**-Low Latency and Efficient Voice Activator for Edge Devices
- **Theme**-Smart Automation
- **PS Category**- Hardware
- **Team ID**- 168867
- **Team Name**- LUMORA6



IDEA TITLE

Proposed Solution

Hybrid Edge-Cloud Voice Activation for Low Latency and Efficient Speech Processing

1 Existing Problem

Cloud-first Voice Activation

Continuous audio is sent to the cloud for processing.



-  **High latency**
Round-trip to cloud
-  **High bandwidth usage**
Continuous audio transmission
-  **Privacy concerns**
Always-on audio sent to cloud
-  **Cloud dependency**
Does not work without internet

2 Our Proposed Solution

Hybrid Edge-Cloud Voice Activation

Detect the custom wake word locally on a low-power device and stream only the relevant audio to a remote ASR server.



3 Innovation and Uniqueness

Beyond the basic SIH requirement, we add intelligence for a more robust and practical system.



Adaptive Two-Stage KWS

Ultra-light first-stage detector with high-confidence verification to reduce false activations.



Intelligent Pre-Roll Buffer

Maintains a short audio buffer before detection to ensure no part of the user's command is missed.



Noise-Adaptive Detection

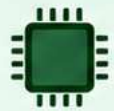
Automatically adjusts processing and detection threshold based on environmental noise.



Always listening locally — **transmitting only when necessary.**

Implementation Methodology

1 Technologies Used



Edge Device
ESP32 / STM32
+ Microphone



Edge AI
MFCC + CNN
(TFLite Micro)

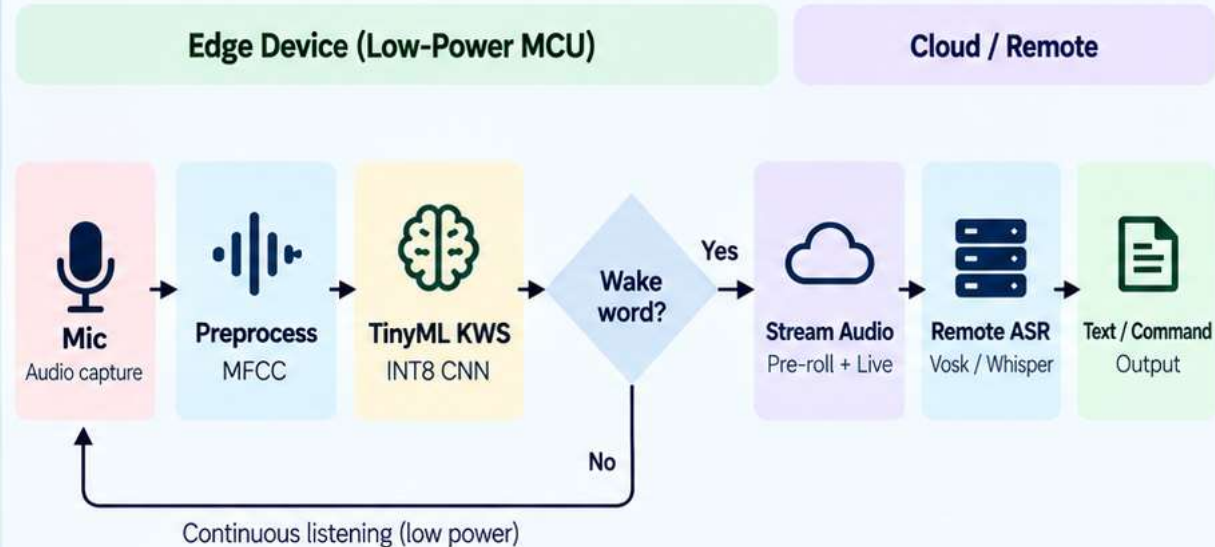


Communication
Wi-Fi / Internet

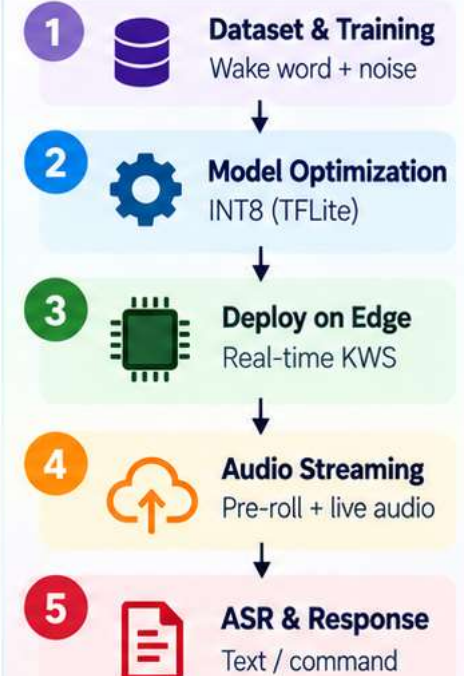


Remote ASR
Vosk / Whisper
(Python)

2 System Block Diagram



3 Implementation Workflow



Innovation & Uniqueness (Implementation Level)



Two-stage processing
(Edge KWS + Cloud ASR)



INT8 optimization
for low power



Adaptive buffer
reduces false activation

Target (SIH Evaluation Criteria)



< 256 KB
RAM



< 10%
Idle CPU



Low latency
(KWS → ASR)

04 FEASIBILITY, CHALLENGES AND MITIGATION STRATEGY

1 FEASIBILITY ANALYSIS



Technical Feasibility

- MFCC + lightweight CNN (KWS)
- Deployable on ESP32 / STM32



Economic Feasibility

- Low-cost hardware (≈ ₹12,000)
- Open-source tools (TFLite, Vosk)



Operational Feasibility

- Real-time, low-power operation
- Continuous local listening with selective cloud streaming



Social Feasibility

- Hands-free and accessible interaction
- Useful in homes, industries and public spaces



Environmental Feasibility

- Low power consumption (edge processing)
- Reduces unnecessary cloud usage

2 POTENTIAL CHALLENGES AND RISKS



False Activation

- Background noise and similar words
- May trigger unintentionally



Resource Constraints

- Limited RAM / CPU on MCU
- Model size and real-time processing load



Network Dependency

- Requires internet for remote ASR
- Latency or disconnection issues



Variability in User Speech

- Different accents and speaking styles
- May reduce recognition accuracy



Environmental Factors

- High ambient noise and reverberation
- Microphone placement and enclosure



Long-term Reliability

- Model drift over time
- Hardware wear and power stability

3 MITIGATION STRATEGIES



Robust Model Training

- Diverse noise dataset
- Confidence threshold & smoothing



Model Optimization

- INT8 quantization (TFLite Micro)
- Lightweight CNN and efficient MFCC



Buffering & Reconnection

- Pre-roll audio buffer
- Automatic reconnection and retry



Dataset Diversification

- Train with multiple accents and users
- Data augmentation (noise, speed, pitch)



Noise Handling

- Noise filtering (e.g., spectral subtraction)
- Adaptive gain control & proper enclosure



Continuous Monitoring

- Periodic model re-training
- Hardware health checks and logging

ESTIMATED COST – EXISTING SOLUTION (COMMERCIAL)



₹ 3,000 – 10,000+ (Per device)

- Commercial wake-word engines (e.g., Picovoice)
- Includes hardware + software licensing cost
- Proprietary / closed-source

VS

ESTIMATED COST – OUR PROPOSED PROTOTYPE



₹ 10,000 – 15,000 (Approximate)

- MCU (ESP32-S3/STM32) + MEMS microphone
- Prototype PCB, enclosure and peripherals
- Open-source software (no licensing cost)



- Lower cost
- No proprietary SDK
- Reduced cloud usage
- Scalable and open-source
- Suitable for real-world deployment

05 IMPACT ON TARGET AUDIENCE AND BENEFITS

1 POTENTIAL IMPACT ON TARGET AUDIENCE



Homes & Offices

- Hands-free and convenient control
- Improved accessibility for all users



Industrial / Commercial Spaces

- Touchless operation in critical environments
- Enables automation and better safety



Elderly & Differently-abled Users

- Easier device interaction
- Enhanced independence and safety



Rural / Low-Connectivity Areas

- Low-cost edge processing
- Works with selective cloud usage



Public Infrastructure

- Useful in smart buildings, hospitals and public spaces
- Reduces manual operation and improves hygiene

2 BENEFITS OF THE SOLUTION



SOCIAL BENEFITS

- Accessible and inclusive interaction
- Improved convenience and safety
- Useful for elderly and differently-abled users



ECONOMIC BENEFITS

- Low-cost hardware (~ ₹12,000 prototype)
- Reduced cloud and licensing costs
- Can be integrated into existing systems
- Scalable for homes, industries and public infrastructure



ENVIRONMENTAL BENEFITS

- Lower data transmission due to edge processing
- Energy-efficient operation on low-power MCU
- Reduces unnecessary cloud usage



TECHNOLOGICAL BENEFITS

- Demonstrates real-time TinyML deployment on edge devices
- Open-source and modular design
- Scalable to multiple use cases
- Builds a foundation for future AI/IoT applications

**OVERALL
IMPACT**



**Greater
Accessibility**



**Safer & Smarter
Environments**



**Sustainable
and Energy-Efficient**



**Cost-Effective
and Scalable**

- **TensorFlow Lite for Microcontrollers**

Lightweight machine-learning framework for deploying neural-network models on resource-constrained microcontrollers.

Reference:

<https://www.tensorflow.org/lite/microcontrollers>

- **openWakeWord**

Open-source framework for wake-word detection and custom model development. Useful as a reference for lightweight keyword spotting and custom wake-word training.

Reference:

<https://github.com/dscripka/openWakeWord>

- **Espressif ESP-SR / WakeNet**

Open-source speech-recognition framework for ESP32 platforms, including local wake-word detection and voice-processing components.

Reference:

<https://github.com/espressif/esp-sr>